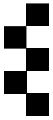
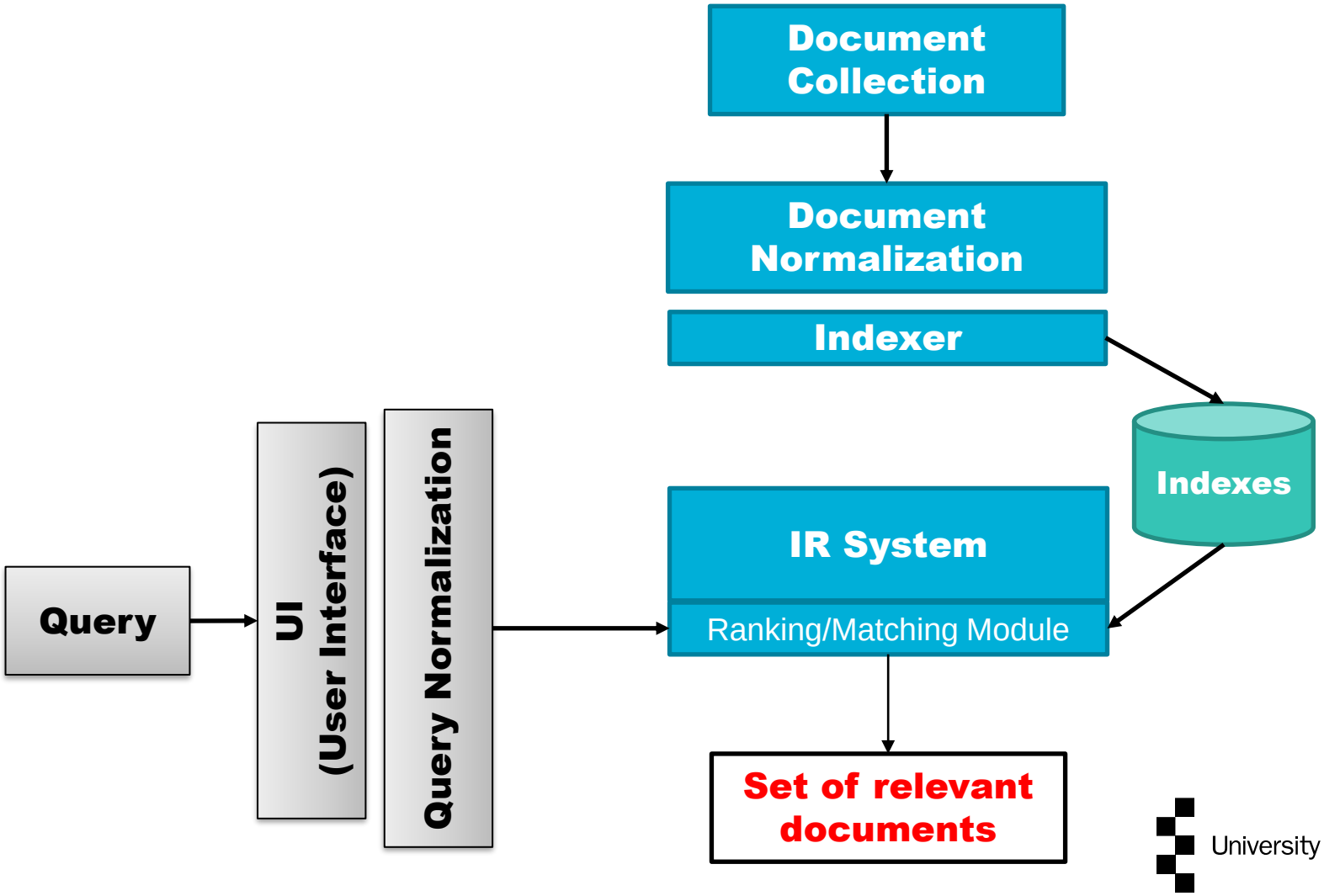
Structure of an IR Model



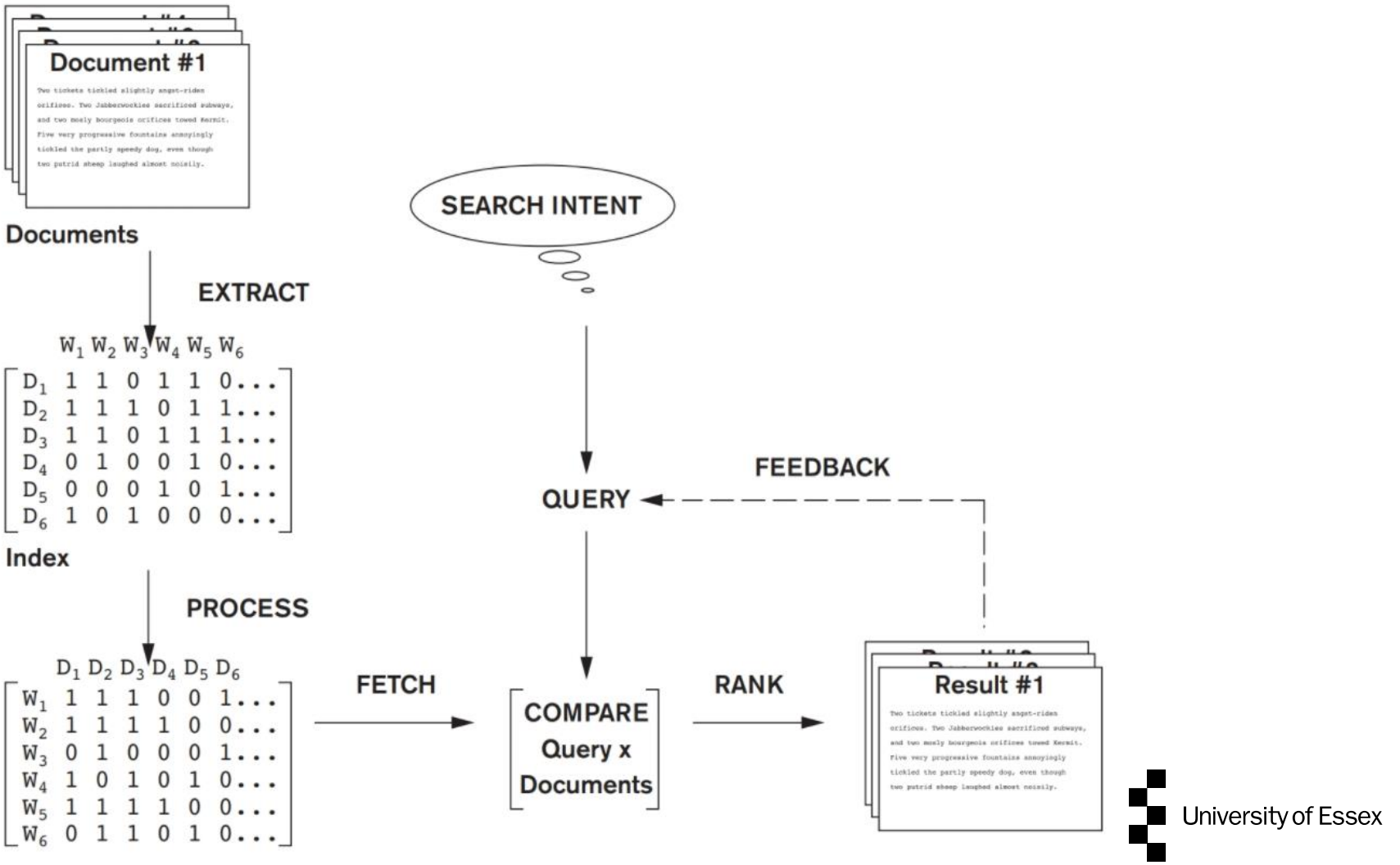
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IR System Components



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Simplified IR Process Pipeline



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Document Pre-processing

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Document/Text Pre-processing

The first task of an IR system is to pre-process the documents.

- Document Granularity
- Tokenization
- Stemming and Lemmatization
- Stop Words, Accents, Case Folding, and Language Identification

Document

- What exactly is a document?
- Is an entire book considered as one document? Or, is each chapter in a book a document?
- consider a chain of email messages.
- Is the entire chain one document or each email in the chain a separate document?
- It is referred as document granularity.
- *The right level of granularity improves the effectiveness of the documentation.*

Tokenization

- A token corresponds to a sequence of bytes.
- For example, in “seven wonders of the world” there are five token types, and each is a different token type.
- On the other hand, in “to be, or not to be” there are **seven** tokens
  - Only **five** distinct token
  - Token “to” and “be” each have two instances
  - Note that “,” is a punctuation token

Tokenization Issues (1)

One or two words?

- Hewlett-Packard
- State-of-the-art
- co-education
- the hold-him-back-and-drag-him-away maneuver
- data base
- San Francisco
- Los Angeles-based company
- cheap San Francisco-Los Angeles fares
- York University vs. New York University

Numbers

20/3/91
3/20/91
Mar 20, 1991
B-52
100.2.86.144
(800) 234-2333
800.234.2333

No whitespace in Chinese Language

莎拉波娃现在居住在美国东南部的佛罗里达。今年4月9日，莎拉波娃在美国第一大城市纽约度过了18岁生日。生日派对上，莎拉波娃露出了甜美的微笑。

Tokenization Issues (2)

Compound Words

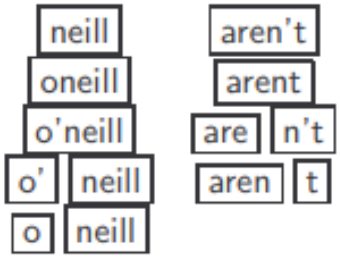
Compounding in Dutch, German, Swedish

German

Lebensversicherungsgesellschaftsangestellter

leben+s+versicherung+s+gesellschaft+s+angestellter

Apostrophes



Ambiguous segmentation

和尚

- As one word, means "monk"
- As two words, means "and" and "still"

Bidirectionality

- Direction of writing changes in some scripts (writing systems); e.g., Arabic.

استقلت الجزائر في سنة 1962 بعد 132 عاما من الاحتلال الفرنسي.

← START ← → ← →

'Algeria achieved its independence in 1962 after 132 years of French occupation.'

Token Normalization

Token Normalization

- refers to canonicalizing tokens so that matches occur despite superficial differences.
- refers to as equivalence classing of tokens.

Examples:

- U.S.A/USA, Los Angles/LA, Louisiana/LA
- 04/01/2018, 2018/04/01, and April 4th, 2018

Lemmatization

- Reduce inflectional/variant forms to base form

am, are, is → be
car, car's, cars', cars → car
the boy's cars are different colours → the boy car be different color

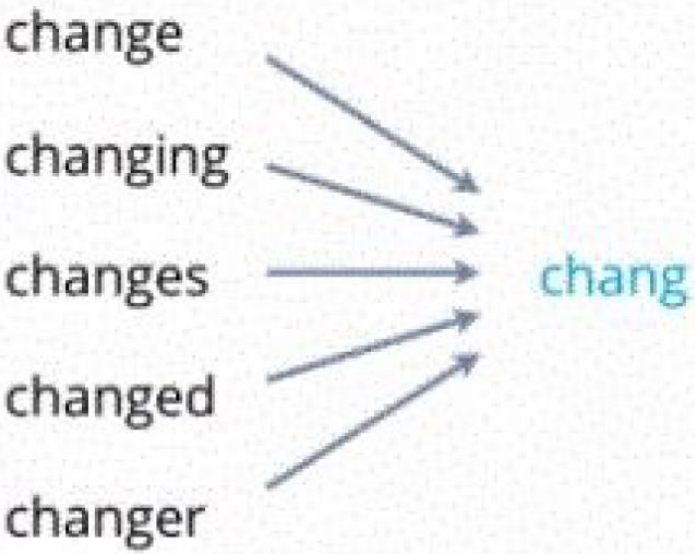
Stemming

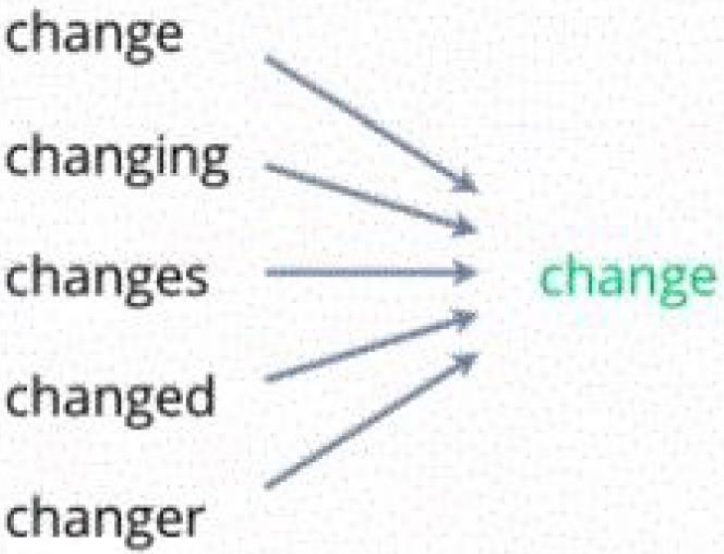
- Stemming is a crude heuristic process which collapses derivationally related words to their stem, base, or root form.

automate, automation, automatic → automat

- There are two kinds of stemmers
  - algorithmic
  - dictionary

Stemming vs Lemmatization





Case Folding

Case folding is another normalization task.

- Reduce all letters to lower case
- Even though case can be semantically distinguishing

Fed vs. fed
March vs. march
Turkey vs. turkey
US vs. us

- Best to reduce to lowercase because users will use lowercase regardness of correct capitalisation.

Stop Words

- Extremely common words which are of little value in helping select documents matching a user need

a, an, and, are, as, at, be, by, for, from, has, he, in, is, it, its, of, on, that, the, to, was, were, will, with

- Used to be standard in older IR systems.
- Need them to search for

to be or not to be
prince of Denmark
bamboo in water

- Length of practically used stoplists has shrunk over the years.
- Most web search engines do index stop words.

Accents and Diacritics

- Accents and diacritics may be ignored in English text, but they can be quite significant for retrieval in other languages such as Spanish.



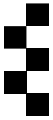


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Language Identification (LID)

- Writing systems of languages also pose problems for token extraction. In some writing systems, one reads from **left to right**, in others, **right to left**, and mixed (**both left to right and right to left**) yet in others.
- The **LID problem** has been solved with **high degree of accuracy** as a classification problem using supervised machine learning approaches.
- However, **mixed-language documents**, where a small fraction of words from another language are mixed, can create **challenges to LID**.

Document Collection and Zipf's Law



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Text Corpus

- Though large IR datasets exist, a small text collection called Mini Gutenberg is created to illustrate select IR concepts.
- Project Gutenberg (2018) provides access to over 56,000 copyright-free or copyright-expired texts.
- In the article, 18 books are download and created a Mini Gutenberg text collection.

Names of documents and other details about the Mini Gutenberg collection

TABLE 1 Mini Gutenberg Text Collection

Doc ID	Document Name	Total Tokens	Token Types
d_1	austen-emma.txt	158,167	14,193
d_2	austen-persuasion.txt	83,308	11,481
d_3	austen-sense.txt	118,675	12,567
d_4	bible-kjv.txt	821,133	25,137
d_5	blake-poems.txt	6,845	3,021
d_6	bryant-stories.txt	45,988	7,669
d_7	burgess-busterbrown.txt	15,870	3,043
d_8	carroll-alice.txt	26,443	5,139
d_9	chesterton-ball.txt	79,543	15,739
d_{10}	chesterton-brown.txt	71,626	15,475
d_{11}	chesterton-thursday.txt	57,955	12,601
d_{12}	edgeworth-parents.txt	166,070	16,639
d_{13}	melville-moby-dick.txt	212,030	33,901
d_{14}	milton-paradise.txt	79,659	17,951
d_{15}	shakespeare-caesar.txt	21,245	5,687
d_{16}	shakespeare-hamlet.txt	29,605	9,399
d_{17}	shakespeare-macbeth.txt	17,741	6,895
d_{18}	whitman-leaves.txt	122,070	24,651
	Total	2,135,242	241,188

Unigrams, Bigrams, and Trigrams

- Unigrams are single words
- bigrams are two-word sequences,
- and trigrams are three-word sequences.

Unigram
the
and
of
to
a
in
i
that
he
it

Bigram
of the
in the
and the
the lord
to the
all the
and he
to be
for the
it was

Trigram
of the lord
the son of
the children of
out of the
the house of
saith the lord
the lord and
and i will
and all the
children of israel

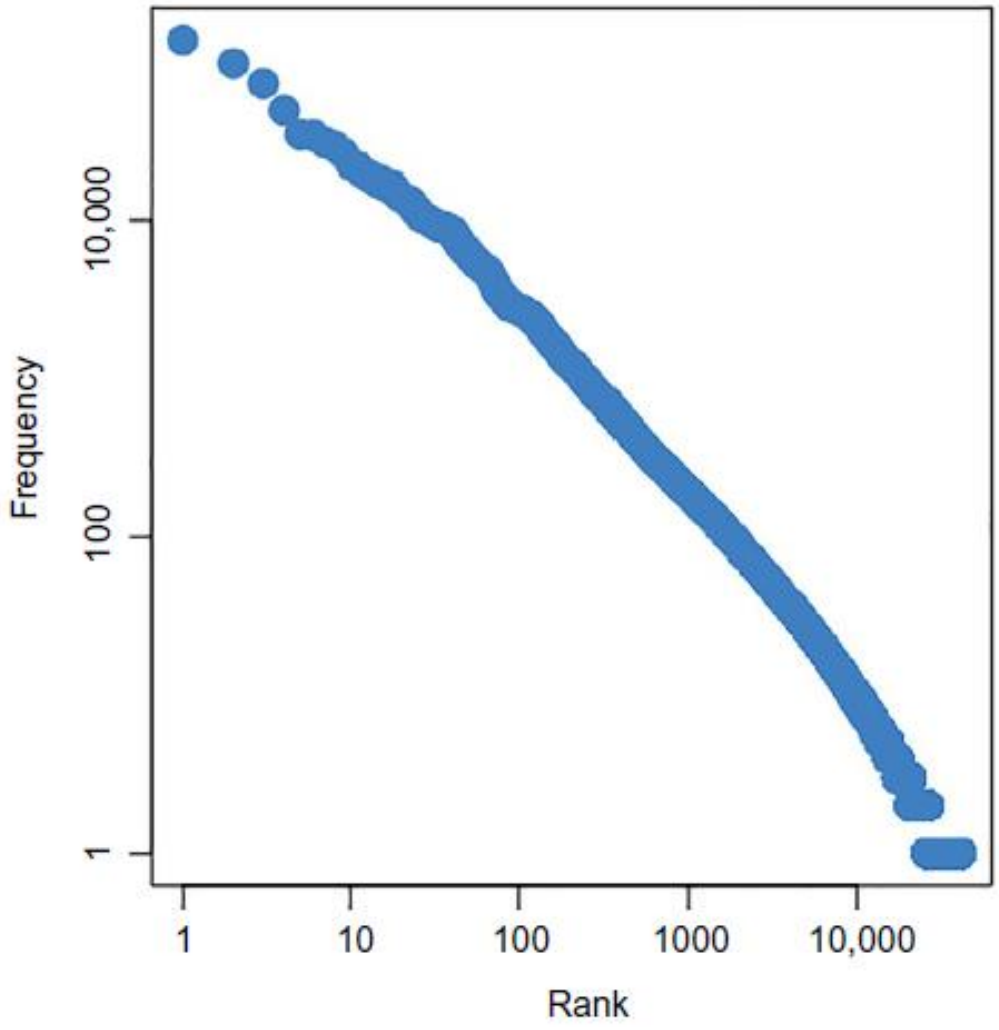
Unigrams in Mini Gutenberg Text Collection and their Frequencies of occurrence

Unigram	Frequency
the	133,603
and	95,444
of	71,269
to	48,060
a	33,962
in	33,580
i	30,300
that	28,811
he	25,868
it	22,304

Zipf's Law for Mini Gutenberg collection

- To determine whether Zipf's law holds for the Mini Gutenberg collection:
- the number of occurrences of words/terms in the collection are counted.
- Next, the words are listed in the decreasing order of their occurrence frequency.
- For a word w_i, let f_i be its frequency of occurrence and r_i be its rank position in the sorted list. $f_i \times r_i$ is equal to some constant k. That is, $f_i \times r_i = k$.
- **-1 slope** corresponds to the Zipf's function.
- We can conclude that the Zipf's law holds reasonably well for the Mini Gutenberg collection.

Log–log plot of word frequencies and their ranks for Gutenberg corpus



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